

Лабораторная Работа №5. Конфигурирование VLAN

Администрирование локальных сетей

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Получить основные навыки по настройке VLAN на коммутаторах сети.

Конфигурация Trunk-порта

```
Password:
msk-donskaya-daborovikov-sw-01>enable
Password:
msk-donskaya-daborovikov-sw-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-01(config)#
msk-donskaya-daborovikov-sw-01#interface g0/1
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-daborovikov-sw-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-01(config)#interface g0/1
msk-donskaya-daborovikov-sw-01(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-01(config-if)#^Z
msk-donskaya-daborovikov-sw-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-01(config)#interface g0/2
msk-donskaya-daborovikov-sw-01(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-01(config-if)#^Z
msk-donskaya-daborovikov-sw-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-01(config)#interface f0/1
msk-donskaya-daborovikov-sw-01(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-01(config-if)#^Z
msk-donskaya-daborovikov-sw-01#write memory
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-01#
```

Figure 1: Конфигурация Trunk-порта msk-donskaya-sw-1

Конфигурация Trunk-порта

```
Password:
msk-donskaya-daborovikov-sw-2>enable
Password:
msk-donskaya-daborovikov-sw-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-2 (config)#interface g0/1
msk-donskaya-daborovikov-sw-2 (config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-2 (config-if)^Z
msk-donskaya-daborovikov-sw-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-2 (config)#interface g0/21^Z
msk-donskaya-daborovikov-sw-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-2 (config)#interface g0/2
msk-donskaya-daborovikov-sw-2 (config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-2 (config-if)^Z
msk-donskaya-daborovikov-sw-2#wr m
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-2#
```

Figure 2: Конфигурация Trunk-порта msk-donskaya-sw-2

Конфигурация Trunk-порта

```
Password:

msk-donskaya-daborovikov-sw-3>enable
Password:
msk-donskaya-daborovikov-sw-3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-daborovikov-sw-3(config)#interface g0/1
msk-donskaya-daborovikov-sw-3(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-3(config-if)#^Z
msk-donskaya-daborovikov-sw-3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-daborovikov-sw-3(config)#^Z
msk-donskaya-daborovikov-sw-3#wr m
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-3#
```

Figure 3: Конфигурация Trunk-порта msk-donskaya-sw-3

Конфигурация Trunk-порта

```
Password:

msk-donskaya-daborovikov-sw-4>enable
Password:
msk-donskaya-daborovikov-sw-4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-daborovikov-sw-4(config)#interface g0/1
msk-donskaya-daborovikov-sw-4(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-4(config-if)#^Z
msk-donskaya-daborovikov-sw-4#wr m
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-4#
```

Figure 4: Конфигурация Trunk-порта msk-donskaya-sw-4

Конфигурация Trunk-порта

```
Password:
msk-pavlovskaya-daborovikov-sw-1>enable
Password:
msk-pavlovskaya-daborovikov-sw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-pavlovskaya-daborovikov-sw-1(config)#interface f0/24
msk-pavlovskaya-daborovikov-sw-1(config-if)#switchport mode trunk

msk-pavlovskaya-daborovikov-sw-1(config-if)#^Z
msk-pavlovskaya-daborovikov-sw-1#wr m
Building configuration...
[OK]
msk-pavlovskaya-daborovikov-sw-1#
```

Figure 5: Конфигурация Trunk-порта msk-pavlovskaya-sw-1

msk-donskaya-sw-1 как VTP-сервер

```
msk-donskaya-daborovikov-sw-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-01(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-daborovikov-sw-01(config)#vtp domain donskaya
Changing VTP domain name from NULL to donskaya
msk-donskaya-daborovikov-sw-01(config)#vtp password cisco
Setting device VLAN database password to cisco
msk-donskaya-daborovikov-sw-01(config)#vlan 2
msk-donskaya-daborovikov-sw-01(config-vlan)#name management
msk-donskaya-daborovikov-sw-01(config-vlan)#vlan 3
msk-donskaya-daborovikov-sw-01(config-vlan)#name servers
msk-donskaya-daborovikov-sw-01(config-vlan)#vlan 101
msk-donskaya-daborovikov-sw-01(config-vlan)#name dk
msk-donskaya-daborovikov-sw-01(config-vlan)#vlan 102
msk-donskaya-daborovikov-sw-01(config-vlan)#name departaments
msk-donskaya-daborovikov-sw-01(config-vlan)#vlan 103
msk-donskaya-daborovikov-sw-01(config-vlan)#name adm
msk-donskaya-daborovikov-sw-01(config-vlan)#vlan 104
msk-donskaya-daborovikov-sw-01(config-vlan)#name other
msk-donskaya-daborovikov-sw-01(config-vlan)#^Z
msk-donskaya-daborovikov-sw-01#wr m
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-01#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
2	management	active	
3	servers	active	
101	dk	active	
102	departaments	active	
103	adm	active	
104	other	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Figure 6: msk-donskaya-sw-1 как VTP-сервер

```
[OK]
msk-donskaya-daborovikov-sw-2#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-daborovikov-sw-2(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-daborovikov-sw-2(config)#interface range f0/1 - 2
msk-donskaya-daborovikov-sw-2(config-if-range)#switchport mode access
msk-donskaya-daborovikov-sw-2(config-if-range)#switchport access vlan 3
msk-donskaya-daborovikov-sw-2(config-if-range)^Z
msk-donskaya-daborovikov-sw-2#wr m
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-2#
```

Figure 7: msk-donskaya-sw-2 как VTP-клиент

```
msk-donskaya-daborovikov-sw-3#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-sw-3(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-daborovikov-sw-3(config)#interface range f0/1
msk-donskaya-daborovikov-sw-3(config-if-range)#switchport mode access
msk-donskaya-daborovikov-sw-3(config-if-range)#switchport access vlan 3
msk-donskaya-daborovikov-sw-3(config-if-range)#^Z
msk-donskaya-daborovikov-sw-3#wr m
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-3#
```

Figure 8: msk-donskaya-sw-3 как VTP-клиент

```
Password:

msk-donskaya-daborovikov-sw-4>enable
Password:
msk-donskaya-daborovikov-sw-4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-daborovikov-sw-4(config)#interface g0/1
msk-donskaya-daborovikov-sw-4(config-if)#switchport mode trunk

msk-donskaya-daborovikov-sw-4(config-if)^Z
msk-donskaya-daborovikov-sw-4#wr m
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-4#conf terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-daborovikov-sw-4(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-daborovikov-sw-4(config)#interface range f0/1 - 5
msk-donskaya-daborovikov-sw-4(config-if-range)#switchport mode access
msk-donskaya-daborovikov-sw-4(config-if-range)#switchport access vlan 101
msk-donskaya-daborovikov-sw-4(config-if-range)#interface range f0/6 - 10
msk-donskaya-daborovikov-sw-4(config-if-range)#switchport mode access
msk-donskaya-daborovikov-sw-4(config-if-range)#switchport access vlan 102
msk-donskaya-daborovikov-sw-4(config-if-range)#interface range f0/11 - 15
msk-donskaya-daborovikov-sw-4(config-if-range)#switchport mode access
msk-donskaya-daborovikov-sw-4(config-if-range)#switchport access vlan 103
msk-donskaya-daborovikov-sw-4(config-if-range)#interface range f0/16 - 24
msk-donskaya-daborovikov-sw-4(config-if-range)#switchport mode access
msk-donskaya-daborovikov-sw-4(config-if-range)#switchport access vlan 104
msk-donskaya-daborovikov-sw-4(config-if-range)^Z
msk-donskaya-daborovikov-sw-4#wr m
Building configuration...
[OK]
msk-donskaya-daborovikov-sw-4#
```

```
msk-pavlovskaya-daborovikov-sw-1#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-daborovikov-sw-1(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-pavlovskaya-daborovikov-sw-1(config)#interface range f0/1 - 15
msk-pavlovskaya-daborovikov-sw(config-if-range)#switchport mode access
msk-pavlovskaya-daborovikov-sw(config-if-range)#switchport access vlan 101
msk-pavlovskaya-daborovikov-sw(config-if-range)#interface range f0/20
msk-pavlovskaya-daborovikov-sw(config-if-range)#switchport mode access
msk-pavlovskaya-daborovikov-sw(config-if-range)#switchport access vlan 104
msk-pavlovskaya-daborovikov-sw(config-if-range)#^Z
msk-pavlovskaya-daborovikov-sw-1#wr m
Building configuration...
[OK]
msk-pavlovskaya-daborovikov-sw-1#
```

Figure 10: msk-pavlovskaya-sw-1 как VTP-клиент

Статический IP-адрес

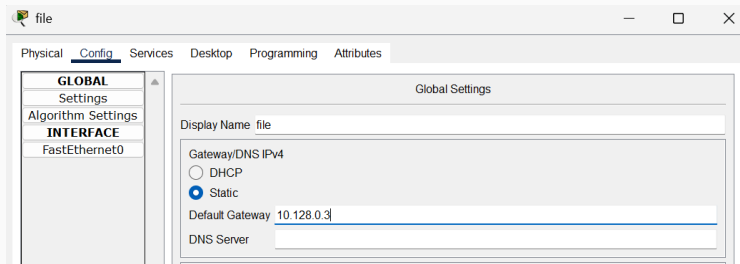


Figure 11: Пример указания статического IP-адреса

Статический IP-адрес

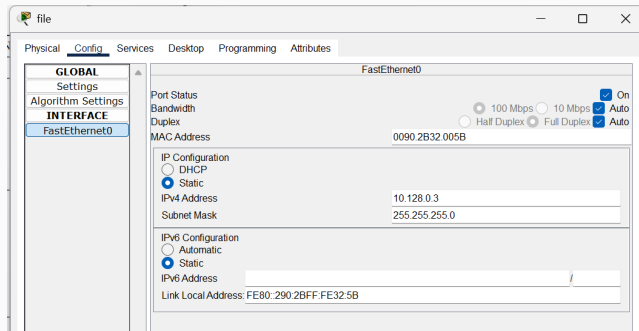
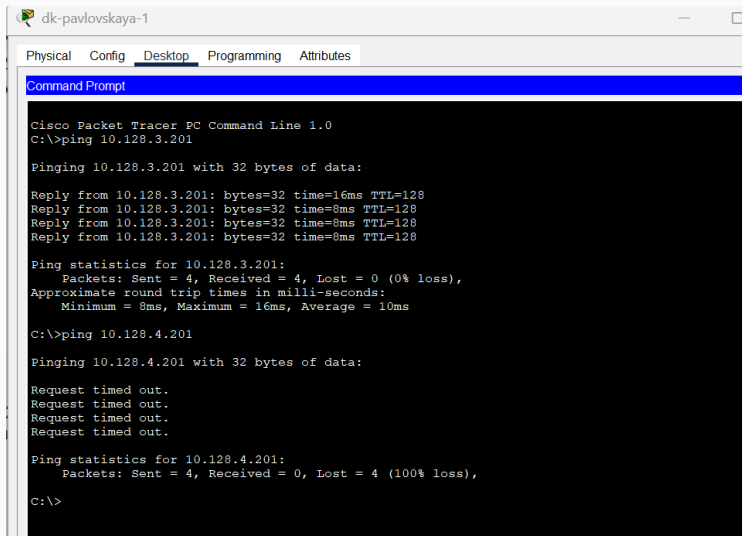


Figure 12: Пример указания статического IP-адреса



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'dk-pavlovskaya-1'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The Command Prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.201

Pinging 10.128.3.201 with 32 bytes of data:

Reply from 10.128.3.201: bytes=32 time=16ms TTL=128
Reply from 10.128.3.201: bytes=32 time=8ms TTL=128
Reply from 10.128.3.201: bytes=32 time=8ms TTL=128
Reply from 10.128.3.201: bytes=32 time=8ms TTL=128

Ping statistics for 10.128.3.201:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 16ms, Average = 10ms

C:\>ping 10.128.4.201

Pinging 10.128.4.201 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.4.201:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 13: ping

Процесс передвижения пакета ICMP

Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	dk-pavlovskaya-1	ICMP
	0.001	dk-pavlovskaya-1	msk-pavlovskaya-daborovikov...	ICMP
	0.002	msk-pavlovskaya-daborovikov-sw-1	msk-donskaya-daborovikov-sw-1	ICMP
	0.003	msk-donskaya-daborovikov-sw-1	msk-donskaya-daborovikov-sw-4	ICMP
	0.004	msk-donskaya-daborovikov-sw-4	dk-donskaya-1	ICMP
	0.005	dk-donskaya-1	msk-donskaya-daborovikov-sw-4	ICMP
	0.006	msk-donskaya-daborovikov-sw-4	msk-donskaya-daborovikov-sw-1	ICMP
	0.007	msk-donskaya-daborovikov-sw-1	msk-pavlovskaya-daborovikov...	ICMP
	0.008	msk-pavlovskaya-daborovikov-sw-1	dk-pavlovskaya-1	ICMP
	1.011	--	dk-pavlovskaya-1	ICMP
	1.012	dk-pavlovskaya-1	msk-pavlovskaya-daborovikov...	ICMP

Figure 14: Процесс передвижения пакета ICMP

Содержимое пакета

PDU Information at Device: msk-donskaya-daborovikov-sw-1



OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: msk-donskaya-daborovikov-sw-1

Source: dk-pavlovskaya-1

Destination: 10.128.3.201

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer 2: Dot1q Header

0060.5C43.ABA6 >> 00D0.FF00.736E

Layer 1: Port GigabitEthernet0/2

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer 2: Dot1q Header

0060.5C43.ABA6 >> 00D0.FF00.736E

Layer 1: Port(s): FastEthernet0/1

1. The frame source MAC address was found in the MAC table of Switch.
2. This is a unicast frame. Switch looks in its MAC table for the destination MAC address.

Challenge Me

<< Previous Layer

Next Layer >>

В ходе выполнения лабораторной работы я получил основные навыки по настройке VLAN на коммутаторах сети